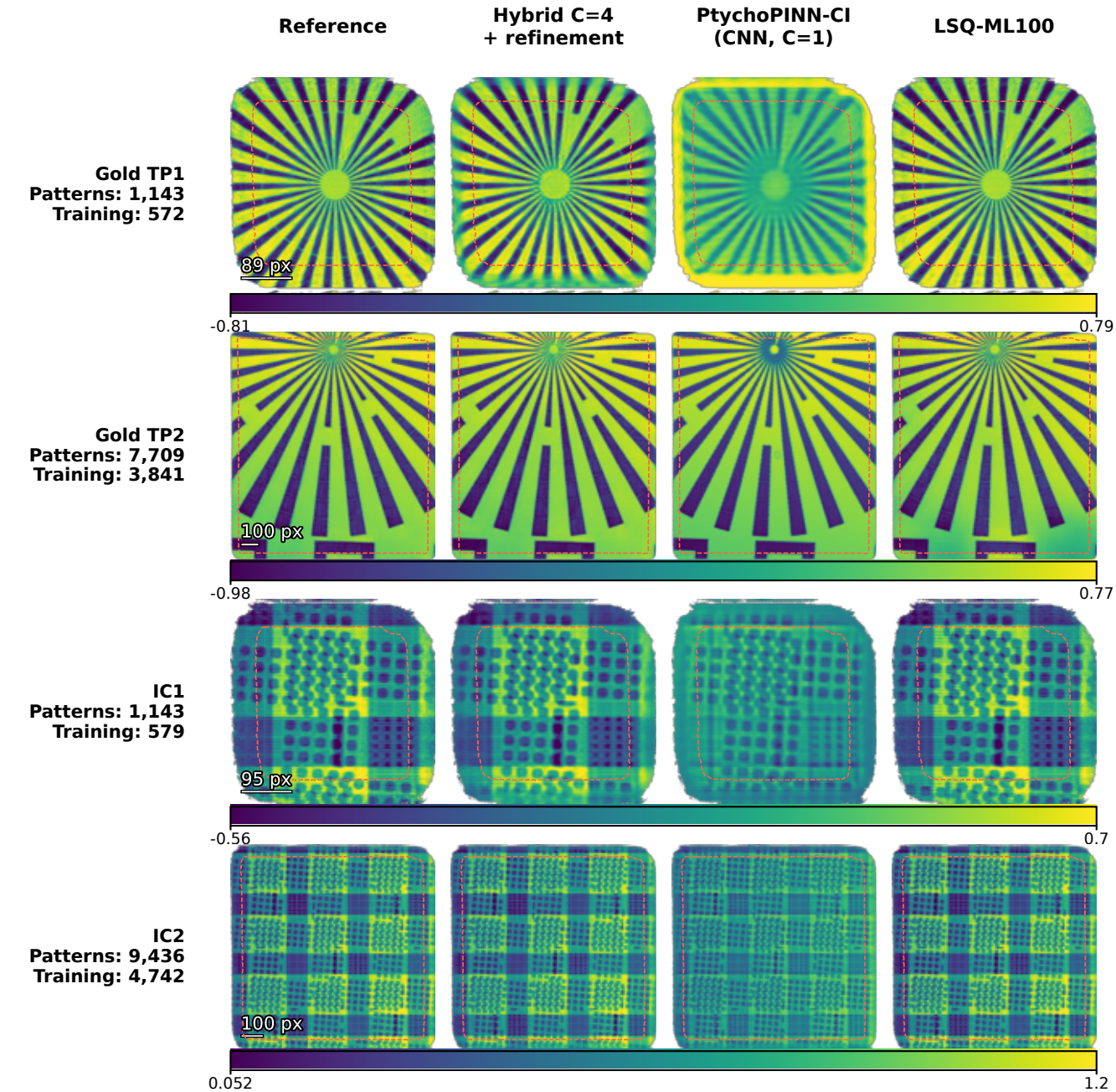
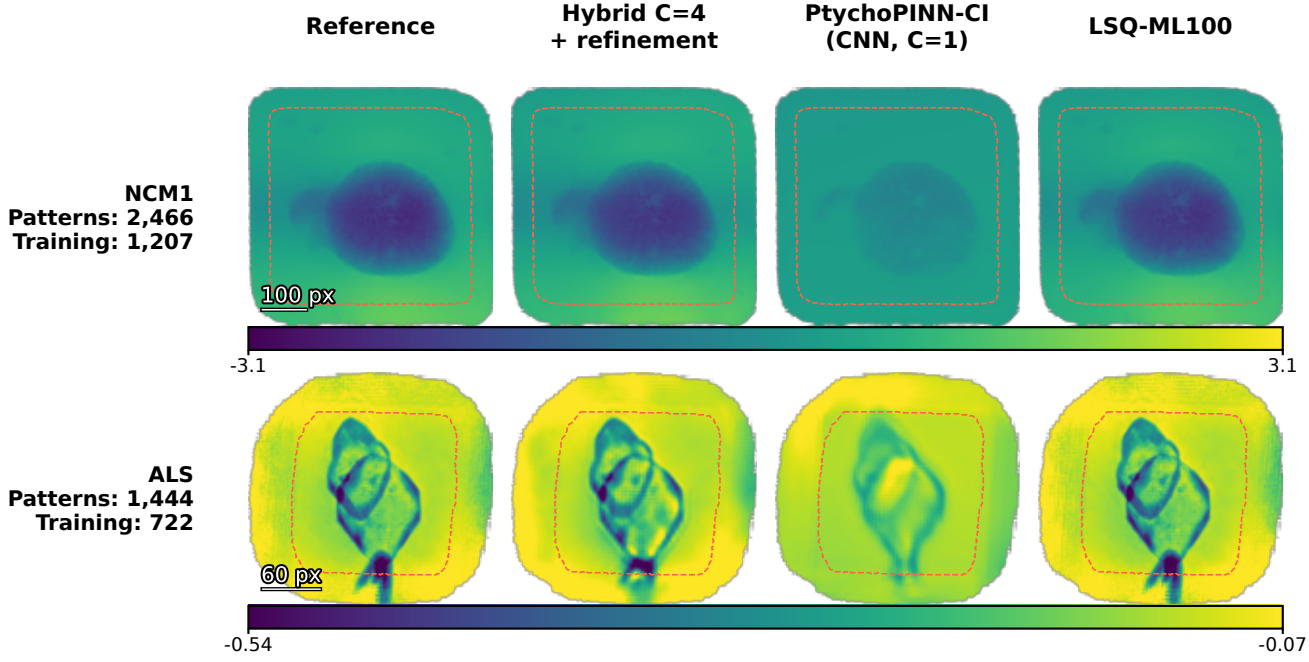


Phase is shown in radians; amplitude shows relative object amplitude. Pattern counts refer to the displayed acquisition and distinct training patterns. High half selects checkpoints (spatial validation, not an independent test; no guard band). Reference is an experimental estimate; dashed contour marks scored interior. Locally trained PtychoPINN-CI (CNN, C=1) uses the same measured training half as Hybrid ResNet. No patch-offset alignment, tilt seeding or phase-field refinement is applied; intrinsic CI calibration and scalar count calibration are retained. This is a local measured-data result, distinct from the published synthetic-only reconstruction. Whole-acquisition images include the training region. Paired groups assign C4 only where it has lower phase MAE after refinement and higher phase SSIM; other datasets use C1. This is the C4-selected group. C4 uses 10 groups per training center (10× oversampling); training compute is not matched to C1. TP2 is shown full field only; FRC is excluded because of scan-grid artefacts.



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